

EDA Interview Questions & Answers

1. Which plots do you use to check correlation?

- Heatmap
- Scatter Plot
- Pairplot

2. How do you handle skewed data?

Methods:

- Log Transformation
- Square Root Transformation
- Box-Cox Transformation

Example:

```
import numpy as np
df['Salary'] = np.log(df['Salary'])
```

3. How to detect multicollinearity?

Methods:

- Correlation Matrix
- Heatmap
- VIF (Variance Inflation Factor)

4. What are Univariate, Bivariate and Multivariate Analysis?

Univariate

One Variable Analysis

Example: Age Distribution

Bivariate

Two Variables Analysis

Example: Age vs Salary

Multivariate

More than Two Variables Analysis

Example: Age, Experience, Salary

5. Difference between Heatmap and Pairplot?

Heatmap:

- Shows correlation values
- Quick overview
- Numeric relationship

Pairplot:

- Shows scatter plots
- Detailed visualization
- Variable distributions

6. How do you summarize your insights?

- Mention key patterns
- Mention correlations
- Mention trends
- Mention outliers
- Give business conclusions

7. What is EDA and Why is it Important?

EDA (Exploratory Data Analysis) is the process of examining a dataset using visual and statistical methods to identify patterns, trends, missing values, outliers, and relationships.

Importance:

- Data Understanding
- Better Feature Selection
- Outlier Detection
- Improved Model Performance
- Better Business Decisions